FLi-SLAM: Algorithm Hardware Codesign for Iterative Closest Point Registration in 3D LiDAR SLAM

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01 BACKGROUND

1 LiDAR measures the scene as 3D points



2 Feature Based Registration Pipeline

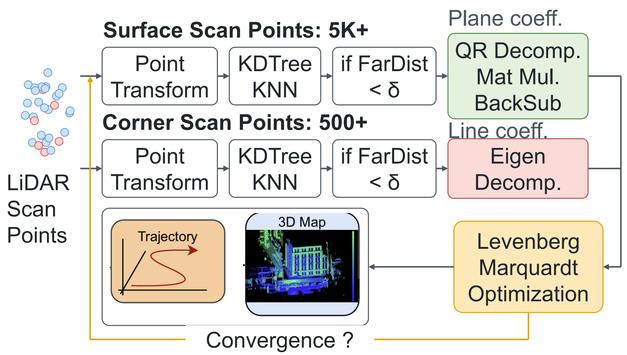
Surface Points: ~100K
Corner Points: ~20K

Map Points Based on Prior Scans

Surface Map Tree
- KDTree Depth: ~17

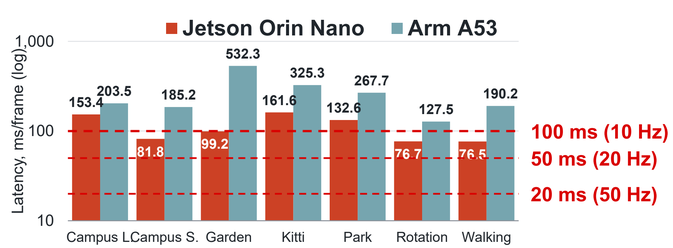
Corner Map Tree
- KDTree Depth: ~15

Build KDTree for Map Points



02 CHALLENGE

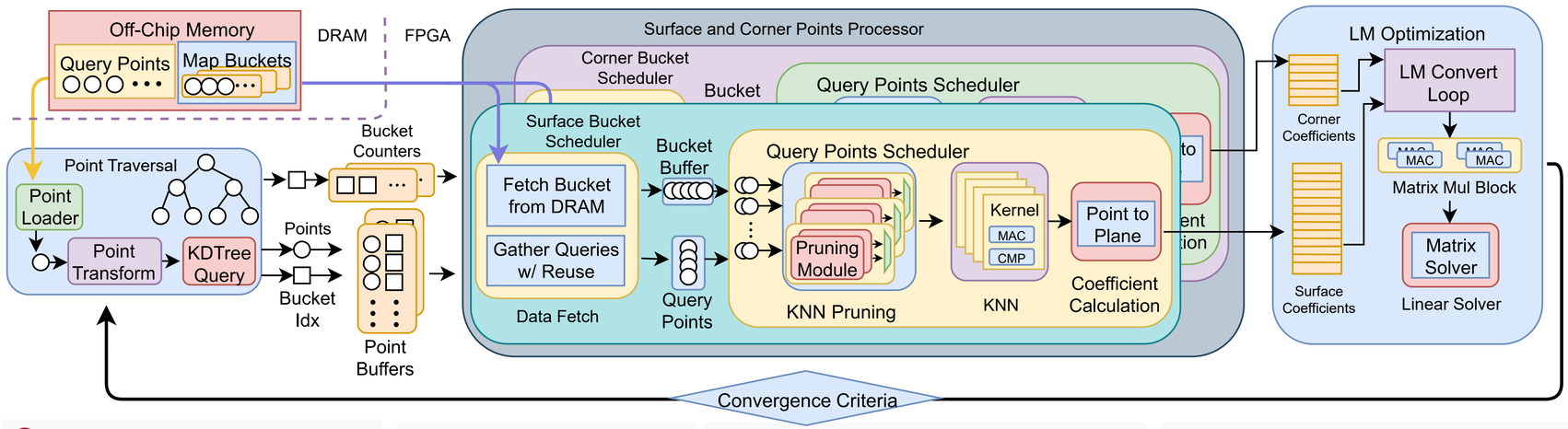
2–5× over the 100 ms required time
original algorithm on ARM A53



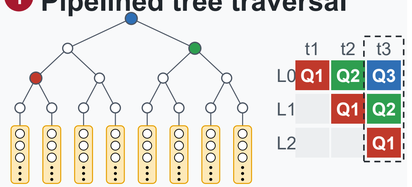
Cause: KNN dominates ICP time

(Surface + Corner) KNN: 87.21%

03 SOLUTION: FIVE CODESIGN INNOVATIONS



1 Pipelined tree traversal



2 Bucket Data Reuse

Queries sharing a bucket reuse one DRAM load.

Without reuse

q1	A	load
q2	B	load
q3	A	load again
q4	C	load
q5	A	load again

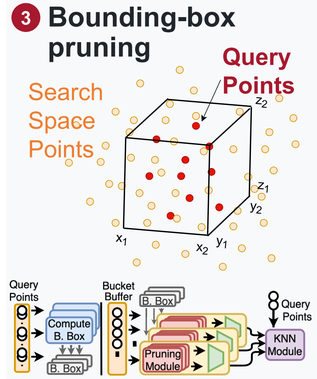
Memory traffic: **A+B+A+C+A**

With bucket reuse

A → q1 q3 q5 load once
B → q2
C → q4

Memory traffic: **A+B+C**

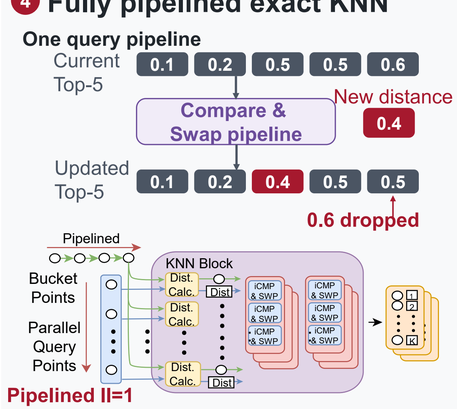
3 Bounding-box pruning



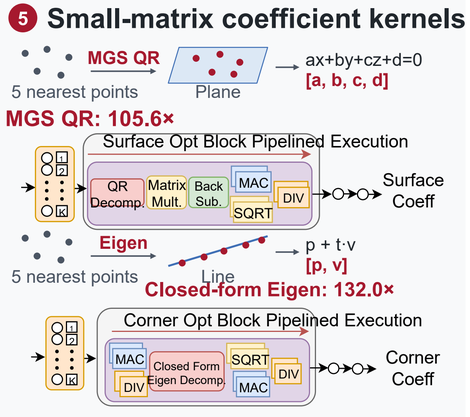
≈86% of bucket points pruned

4 Fully pipelined exact KNN

One query pipeline



5 Small-matrix coefficient kernels



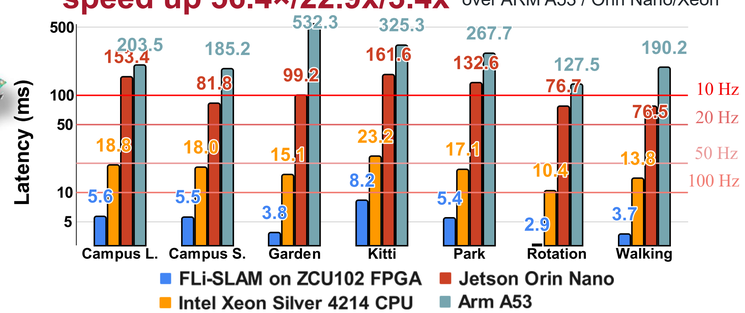
04 RESULT

AMD XILINX



AMD Zynq UltraScale+™ MPSoC ZCU102 Evaluation Kit

2.9–8.2 ms per frame
speed up 56.4×/22.9x/3.4x over ARM A53 / Orin Nano/Xeon



Energy Efficiency Improvement
19.5x, 10.8x and 51.0x over ARM A53 / Orin Nano/Xeon

